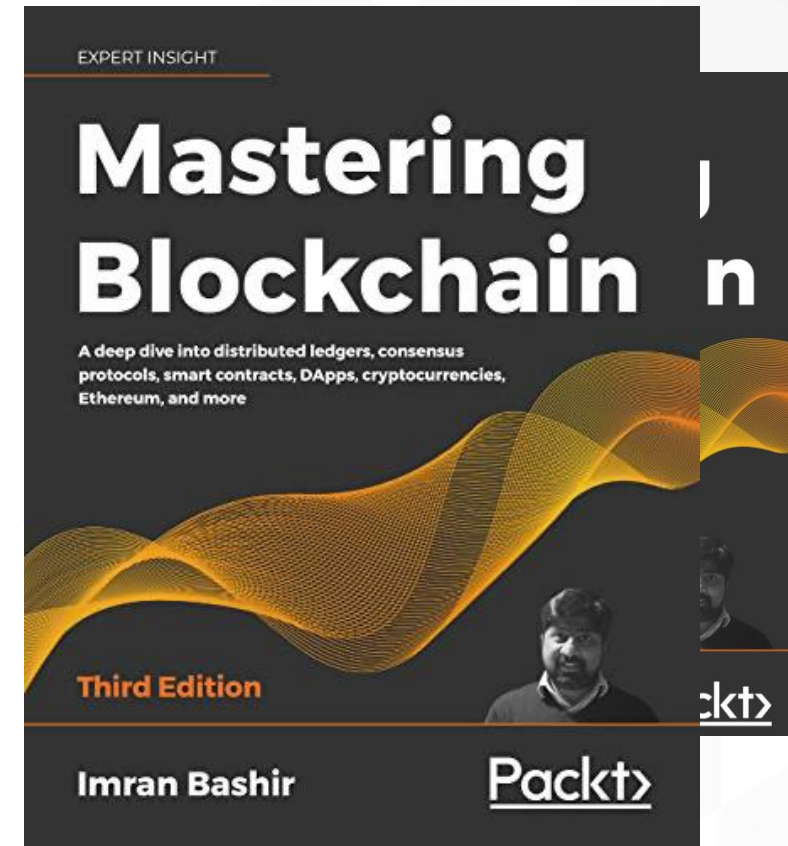


Mastering Blockchain

Third Edition

Chapter 13, Ethereum Development
Environment



Outline



- Test networks
- Private network components
- Starting up the private network
- Mining on the private network
- Remix IDE
- MetaMask
- Using MetaMask and Remix IDE to deploy a smart contract

Test networks



- `--testnet` Ropsten network: pre-configured Proof of Work test network
- `--rinkeby` Rinkeby network: pre-configured Proof of Authority test network
- `--goerli` Görli network: pre-configured Proof of Authority test network

Create a private net



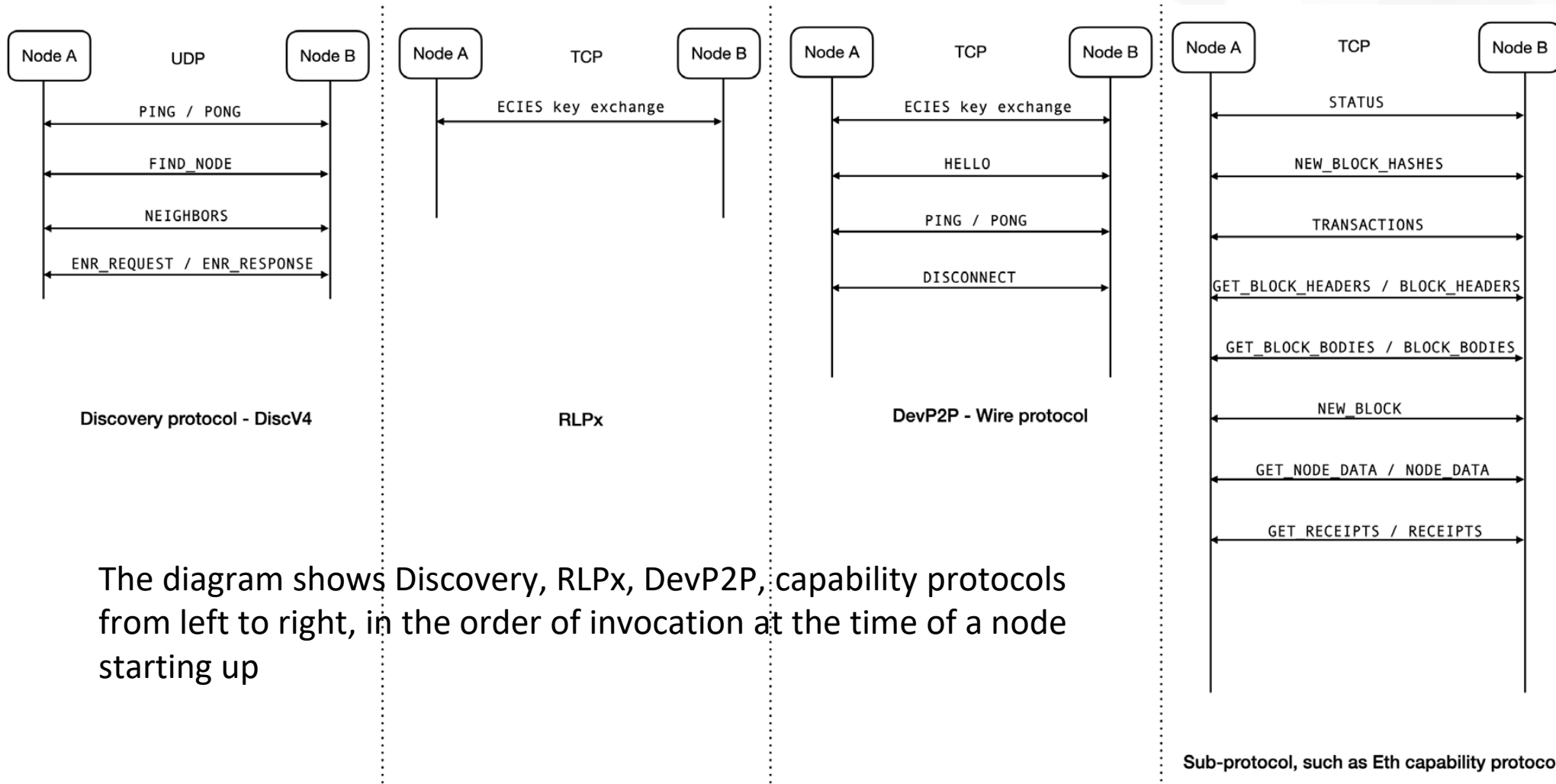
In order to create a private net, three components are needed:

- Network ID
- The genesis file
- A data directory to store blockchain data

Ethereum node discovery

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The Ethereum network stack consists of several protocols, including Discovery, RLPx, DevP2P, and capability protocols



The diagram shows Discovery, RLPx, DevP2P, capability protocols from left to right, in the order of invocation at the time of a node starting up

Genesis file

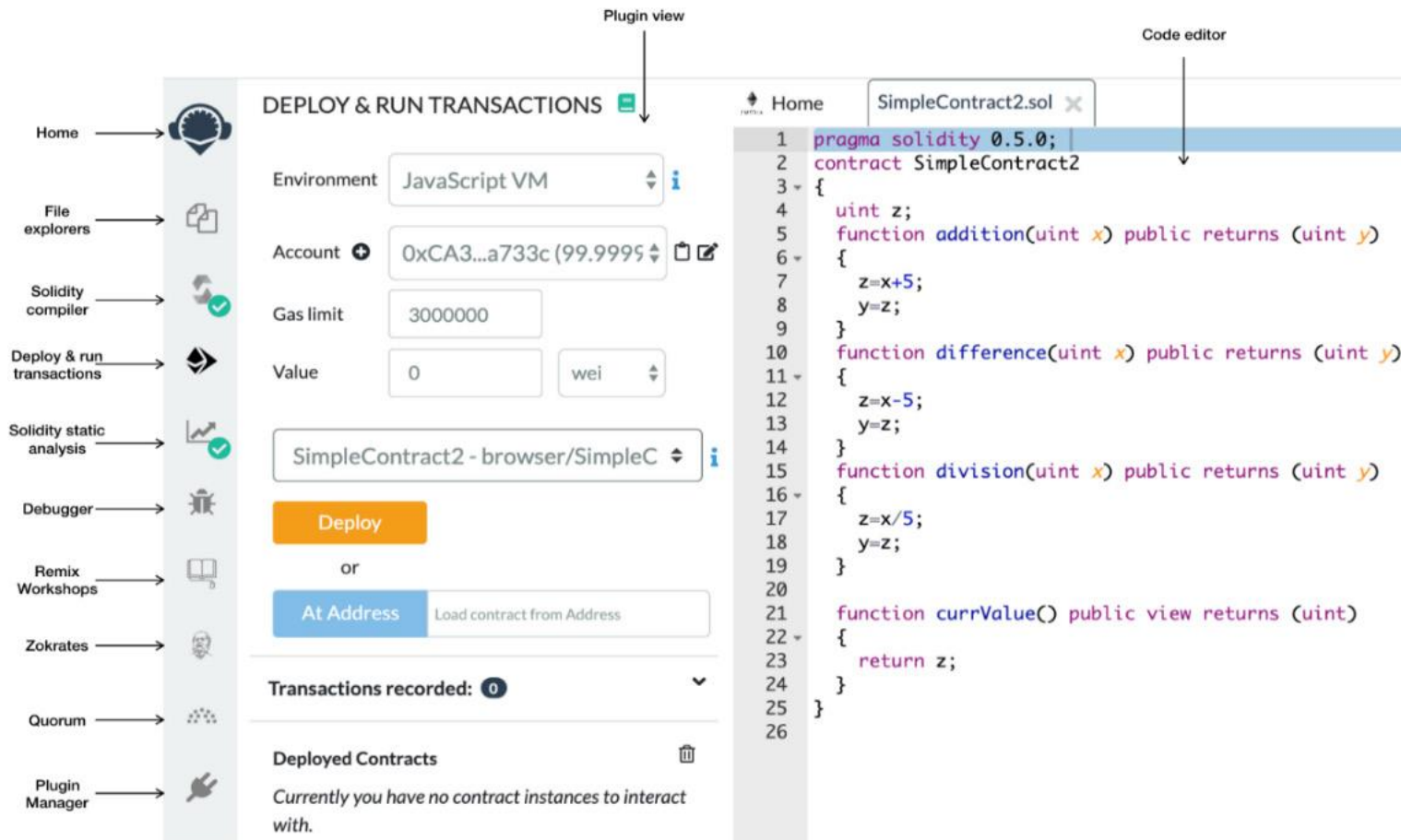
As follows is the Ethereum blockchain genesis file, which defines the chain configuration

```
{
  "nonce": "0x0000000000000042",
  "timestamp": "0x00",
  "parentHash": "0x0000000000000000000000000000000000000000000000000000000000000000",
  "extraData": "0x00",
  "gasLimit": "0x8000000",
  "difficulty": "0x0400",
  "mixhash": "0x0000000000000000000000000000000000000000000000000000000000000000",
  "coinbase": "0x3333333333333333333333333333333333333333",
  "alloc": {
  },
  "config": {
    "chainId": 786,
    "homesteadBlock": 0,
    "eip150Block": 0,
    "eip155Block": 0,
    "eip158Block": 0
  }
}
```

Remix IDE

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Remix IDE is an Ethereum contract development environment, which supports several plugins and has many useful features



Remix IDE debugger

Remix IDE's debugger is a useful debugging tool, which shows a disassembly of the contract code and allows a user to step through the code to aid the debugging process

DEBUGGER 

0x5800c5b1472ac75e5453d47f0f9cf826e...

Stop debugging

136 ADD
137 SWAP1
138 SWAP3
139 SWAP2
140 SWAP1
141 POP
142 POP
143 POP

vm trace step: 99
execution step: 99
add memory:
gas: 0
remaining gas: 2957213

loaded address: 0x08970fed061e7747cd9a38
d680a601510cb659fb

► Solidity Locals 

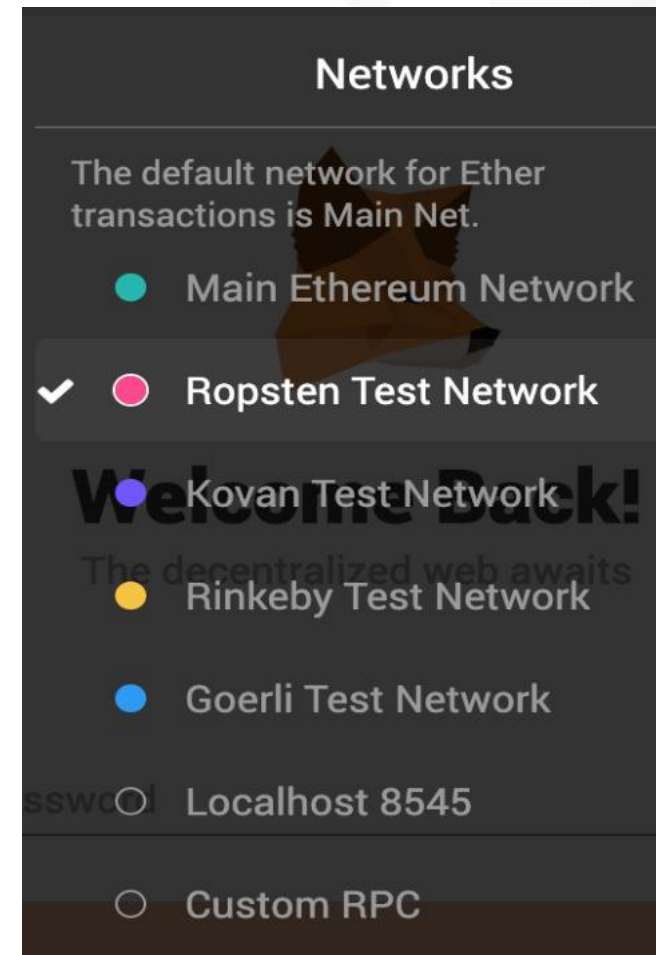
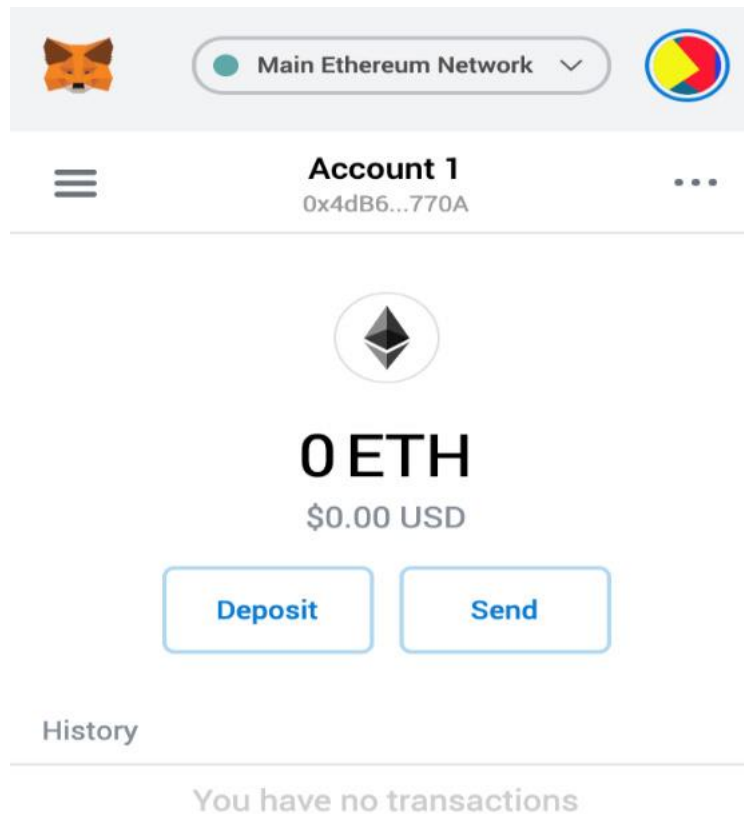
▼ Solidity State 
x: 10 uint8

▼ Stack 
0: 0x00

MetaMask



MetaMask works as a cryptocurrency wallet (as shown in the left image) and can connect with different Ethereum networks (as shown in the right image)



Exercise



In your own time, run `simplecontract2` in the Remix IDE, following the instructions in the *Mastering Blockchain* core book.

Summary

In this presentation, we considered:

- Test networks
- Private network components
- Starting up the private network
- Mining on the private network
- Remix IDE
- MetaMask
- How to use MetaMask and Remix IDE to deploy a smart contract